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Automatic Bottle Filling System Using PLC/SCADA

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ABSTRACT

This project presents an Automatic Bottle Filling System using Siemens S7-1200 PLC and SCADA for real-time monitoring. It uses an IR sensor to detect bottles, an ultrasonic sensor to check liquid levels, and a mini pump to fill the bottles. The conveyor is controlled using DC motors and relays, with logic programmed via TIA Portal V16. The system improves accuracy, reduces manual effort, and is suitable for small to medium-scale industries like beverages and dairy. It provides an efficient, automated solution for modern production lines.

Keywords: Automation, PLC, SCADA, Bottle Filling System, Siemens S7-1200, TIA Portal, IR Sensor, Ultrasonic Sensor, Conveyor Mechanism, Industrial Control, Mini Water Pump.

INTRODUCTION

In today's fast-paced industrial world, automation plays a crucial role in improving productivity, reducing manual errors, and ensuring consistency in manufacturing processes. One such widely used automation application is the automatic bottle filling system, which is highly relevant in industries such as food and beverages, pharmaceuticals, chemicals, and dairy. Manual bottle filling processes are often slow, inaccu- rate, and prone to spillage, leading to product wastage and labor fatigue. To overcome these challenges, this project focuses on designing and implementing an Automatic Bottle Filling System using Programmable Logic Controller (PLC) and Supervisory Control and Data Acquisition (SCADA). The proposed system uses a Siemens S7-1200 PLC programmed using TIA Portal V16, enabling precise control over the sequence of operations. An IR sensor is used to detect the presence of a bottle on the conveyor, while an ultrasonic sen- sor monitors the liquid level inside the bottle. Once the bottle is in position, the conveyor halts, and a mini water pump is activated to fill the bottle. After reaching the desired level, the pump stops, and the conveyor resumes, moving the filled bottle forward. The entire system is monitored through a SCADA interface that allows real-time observation, parameter control, and status feedback. This integration ensures efficiency, reduces manpower, and enhances accuracy in filling operations. The system is cost-effective, scalable, and designed to suit small to medium-scale production units. Overall, it demonstrates how industrial automa- tion can enhance production quality while

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minimizing human involvement and operational delays.

PROBLEM STATEMENT

Manual bottle filling in small and medium industries often leads to issues like uneven filling, spillage, slow processing, and higher labor costs. These challenges affect productivity, accuracy, and hygiene. Without automation, scaling up operations or ensuring real-time monitoring becomes difficult. To overcome these limitations, there is a need for an efficient, automated system that can handle bottle detection, filling, and monitoring with minimal human intervention.

- Manual filling processes are slow and inefficient.
- Inconsistent fill levels and frequent spillage occur due to human error.
- Labor-intensive operations increase production cost.
- Lack of precision reduces overall product quality.
- Real-time monitoring and control are not possible in manual systems.
- Difficult to scale up production in manual environments.

There is a need for an automated system that improves accuracy, efficiency, and hygiene in the bottle filling process.

Solution Statement

To address the limitations of manual bottle filling, this project proposes an automated bottle filling system using a Siemens S7-1200 PLC integrated with SCADA for real-time monitoring and control. The system uses an IR sensor to detect the presence of bottles on the conveyor and an ultrasonic sensor to monitor the liquid level during filling. A mini water pump is used for dispensing the liquid, while a conveyor mechanism con- trolled by DC motors moves the bottles through the process. All components are connected to the PLC, which is programmed using TIA Portal V16 to execute the filling sequence with precision. The SCADA interface enables visualization, live status updates, and manual overrides if necessary. This automated setup ensures consistent fill levels, reduces human effort, improves hygiene, and increases production speed—making it an ideal solution for small and medium-scale industries.

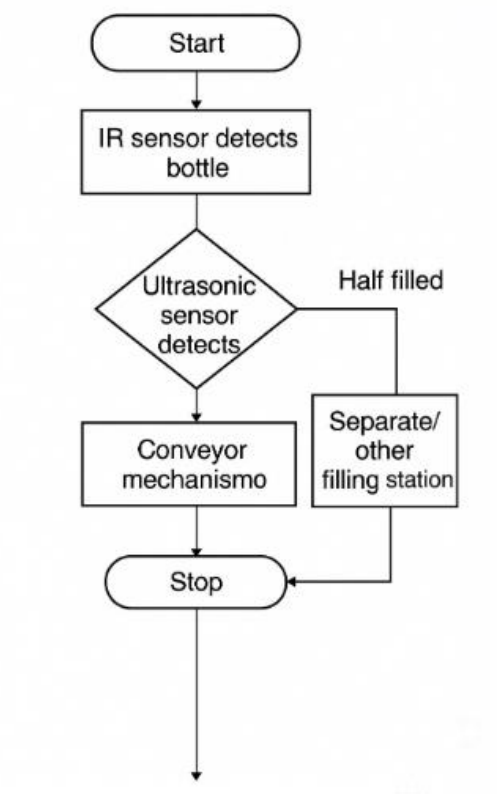

Figure 1. System Flow Diagarm

Methodology

The methodology of this project involves designing and implementing an automated bottle filling system by integrating sensors, actuators, and a PLC-based control unit. First, the components such as the IR sensor, ultrasonic sensor, mini water pump, and conveyor mechanism are selected and connected to the Siemens S7-1200 PLC. The system is powered through a suitable SMPS, and all connections are established following standard wiring practices. Using TIA Portal V16, a ladder logic program is developed to control the sequence of operations—starting from bottle detection, halting the conveyor, initiating the filling process, checking the fill level, and restarting

the conveyor. The SCADA interface is configured to visualize the entire process, display real-time data, and provide manual controls for emergency stop or maintenance purposes. The system is tested and refined through multiple iterations to ensure smooth and reliable performance. This step-by-step approach ensures accurate, fast, and automated bottle filling with minimal human intervention.

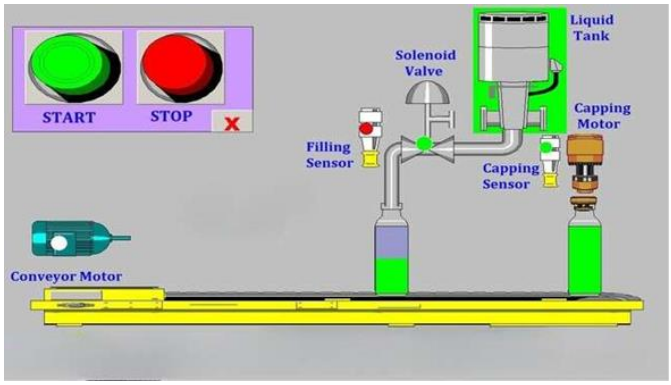

Figure 2. Reference Diagram

The diagram illustrates a basic automated bottle filling and capping system controlled through a PLC in- terface. It begins with a conveyor motor that moves empty bottles toward the filling station. Once a bottle reaches the filling sensor, the conveyor stops, and a solenoid valve opens to dispense liquid from the liq- uid tank into the bottle. After filling, the conveyor restarts, moving the bottle to the next station where the capping sensor detects its presence. The capping motor is then activated to place a cap on the filled bot tle. The entire process is monitored and controlled via an HMI (Human Machine Interface) panel featuring START and STOP push buttons, allowing operators to manually control the system as needed. This setup ensures precise, automated handling of bottles with minimal manual effort, improving speed and accuracy in production.

Hardware

The hardware used in the automated bottle filling and capping system consists of various electromechanical and electronic components integrated through a PLC for precise control and coordination. These include sensors to detect bottle position and level, motors to operate the conveyor and capping unit, a solenoid valve to control liquid flow, and a mini water pump for consistent filling. Each hardware element plays a crucial role in ensuring smooth, automated operation of the entire process.

Table 1. Components and its Specification

Sr No.	Components	Specification
1	PLC (Seimens s7-1200)	Controls entire automation Process
2	IR Sensor	Detects Bottle
3	Ultrasonic Sensor	Checks bottle half- filled or fully filled
4	Solenoid Valve	Controls flow of liquid from tank
5	Mini Water Pump	Pumps liquid from tank to bottle
6	Conveyor Belt	Moves bottle to various stations

The automated bottle filling and capping system relies on a sophisticated integration of electromechani- cal and electronic components, primarily controlled by a Programmable Logic Controller (PLC). This inte- gration ensures precise control and coordination throughout the process, facilitating smooth and efficient operation.

Sensors are crucial components in this system, responsible for detecting bottle position and liquid level. These sensors, often infrared or ultrasonic, provide real-time feedback to the PLC, allowing it to adjust op- erations based on the presence and alignment of bottles, as well as the volume of liquid dispensed. This feedback loop is essential for maintaining accuracy in both filling and capping operations.

Motors play a vital role in driving the conveyor belt and operating the capping unit. Typically, DC motors are used to control the movement of bottles along the

conveyor and to tighten caps during the capping process. The speed of these motors can be adjusted via PLC programming to match specific production requirements, ensuring flexibility and adaptability in the system.

The solenoid valve is another key component, responsible for regulating liquid flow during the filling pro- cess. It operates by opening or closing based on signals from the PLC, ensuring consistent liquid dispensing without overflow. This valve is particularly effective for handling liquids with varying viscosities, making it versatile for different types of products.

A mini water pump is employed to maintain steady liquid flow during filling. This pump works in conjunc- tion with the solenoid valve to deliver precise volumes of liquid into each bottle, ensuring uniformity across batches. The combination of the pump and solenoid valve allows for precise control over the filling process, which is critical for maintaining product quality.

The conveyor system facilitates bottle movement through different stages of filling and capping. Driven by DC motors, it ensures smooth transitions between stations while maintaining alignment for accurate oper- ations. The conveyor's design and operation are optimized to prevent bottle jamming and misalignment, further enhancing the efficiency of the system.

At the heart of the system is the PLC, which serves as the central controller. It processes input signals from sensors, executes control algorithms, and sends output commands to actuators, motors, and valves. The use of PLC enhances system flexibility, allowing easy modifications to operational parameters such as filling volume or conveyor speed. This flexibility is invaluable for adapting to changing production needs or product specifications.

Overall, the hardware components in an automated bottle filling and capping system are intricately inter- connected within the system architecture to ensure efficient automation, minimizing human intervention while maximizing production accuracy and speed.

Figure 3. SEIMENS PLC S7-1200

Figure 4. Hardware

Software

The automated bottle filling system can be effectively developed and simulated using TIA Portal V13, a com- prehensive software environment designed for programming, simulating, and monitoring PLC-based au- tomation systems. TIA Portal V13 supports the Siemens S7-1200 PLC, which is widely used for industrial automation projects due to its reliability and advanced features. This software enables the creation of ladder logic programs, integration of sensors and actuators, and simulation of the entire process without requiring physical hardware.

In this project, the bottle filling system is programmed to detect bottle size, control liquid

dispensing, and co- ordinate conveyor movements. The process begins by defining the inputs and outputs in TIA Portal. Inputs include sensors for detecting bottle presence and size, while outputs control the conveyor motor, solenoid valve for liquid flow, and other actuators involved in the process. For example, height sensors are used to distinguish between small (200 ml) and large (500 ml) bottles. These sensors generate binary data (0 for small bottles and 1 for large bottles), which is stored in a shift register.

The program logic involves tracking bottles using a shift register. Each filling station corresponds to a data bit in the register. As bottles move along the conveyor, their size information is shifted through the register. When a bottle reaches the filling station, the program checks its corresponding bit value: a ”1” triggers long filling (e.g., 4 seconds), while ”0” triggers short filling (e.g., 2 seconds). The shift operation continues until all stations are filled. After completing the filling cycle for eight bottles (based on an 8-bit register), the system resets the register and starts a new cycle.

To implement this in TIA Portal V13:

Define Inputs/Outputs: Configure sensors as inputs (e.g., I0.0 for bottle detection) and actuators as outputs (e.g., Q0.0 for conveyor control).

Create Program Logic: Use ladder diagrams to implement shift operations, timers, counters, and conditional statements for controlling filling durations based on bottle size.

Simulation: Utilize the integrated PLC simulator to test program functionality without physical hardware.

Optimization: Adjust timing parameters (e.g., conveyor speed or filling duration) based on simulation re- sults to ensure smooth operation.

Deployment: Once verified through simulation, deploy the program to an actual S7-1200 PLC for real-world application.

The use of TIA Portal V13 simplifies complex programming tasks with its intuitive interface and powerful simulation tools, making it ideal for developing automated systems like bottle filling machines.

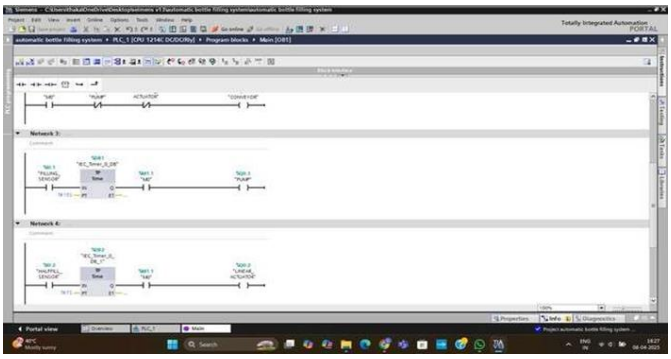

Figure 4. TIA PORTAL V13 LADDER DIAGRAM

Advantages and Disadvantages

The automated bottle filling and capping system offers several advantages, making it an efficient solution for industrial production. One of the primary benefits is its ability to significantly improve productivity by re- ducing manual intervention. Automation ensures consistent filling and capping, minimizing human errors and maintaining uniformity across batches. Additionally, the integration of sensors and PLC-based control systems enhances precision, ensuring accurate liquid dispensing and proper capping. This system also re- duces labor costs and increases operational speed, making it ideal for large-scale production environments. Furthermore, automation contributes to better hygiene standards by limiting direct contact with bottles, which is particularly advantageous in industries like food and beverage or pharmaceuticals.

However, the system has certain disadvantages that must be considered. The initial setup cost of such auto- mated systems can be high due to the need for advanced hardware components like sensors, motors, solenoid valves, and PLCs. Maintenance can also be challenging, as any malfunction in the hardware or software may require specialized expertise to resolve. Additionally, the system’s reliance on electricity

makes it vulnerable to power outages, which can disrupt production schedules. For smaller-scale operations or businesses with limited budgets, the high cost and complexity of automation may not be economically viable. These factors highlight the importance of assessing operational requirements before implementing such systems.

Future Scope

The automated bottle filling and capping system holds significant potential for future advancements, particularly with the integration of Industry 4.0 technologies. By incorporating IoT (Internet of Things) and real-time data analytics, the system can enable predictive maintenance, remote monitoring, and enhanced process optimization. Machine learning algorithms could further improve efficiency by dynamically ad- justing filling and capping parameters based on product variations or production demands. Additionally, advancements in robotics and sensor technology could allow for greater flexibility, enabling the system to handle a wider range of bottle sizes and shapes with minimal reconfiguration. These innovations will make the system more intelligent, adaptive, and suitable for diverse industrial applications.

Conclusion

The automated bottle filling and capping system represents a significant step forward in industrial automation, offering enhanced efficiency, precision, and scalability for production processes. By leveraging advanced hardware components and PLC-based control through software like TIA Portal V13, the system minimizes human intervention while ensuring consistent quality and hygiene standards. Despite challenges such as high initial costs and maintenance requirements, its benefits outweigh the drawbacks, making it a valuable solution for industries aiming to optimize operations. With future advancements in IoT, machine learning, and robotics, this system has the potential to evolve into a more intelligent and adaptive technol- ogy, further revolutionizing automated manufacturing processes.

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